

Econ 120B: Midterm Review

Topics:

Math Review

Definitions

Potential outcomes / Selection bias

Hypothesis Testing

Linear Regression

1. Variance of Y : $Var(Y) = \mathbb{E}[(y - \mu_y)^2] = \sigma_y^2$.

Variance tells us how far our observations of Y are from their expected value

2. Covariance between Z and X : $Cov(Z, X) = \mathbb{E}[(Z - \mu_z)(X - \mu_x)] = \sigma_{zx}$

Covariance tells us to what extent Z and X move together.

Notes:

Also need to know the sample versions and related concepts. Ex:

$\bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i$: sample mean

$s_Y^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2$: sample variance

$s_Y = \sqrt{s_Y^2}$: sample standard deviation

For sample mean, the sample $SE(\bar{Y}) = \sqrt{\frac{s_Y^2}{N}} = \frac{s_Y}{\sqrt{N}}$, sample standard error
of differences in sample means, $SE(\bar{Y}_1 - \bar{Y}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Definitions

1. Central Limit Theorem (CLT): If 1) $Y_1, \dots, Y_n$ are iid, 2) $0 < \sigma_Y^2 < \infty$, and 3) sample size n is large, then $\bar{Y} \sim N(\mu_Y, \sigma_Y^2/n)$, where $\sigma_Y^2 = \sigma_Y^2/n$. Note: CLT $\implies$ we can use the normal distribution for hypothesis testing if n is large.

2. Unbiased estimator: An estimator of a population parameter ($\hat{\theta}$) is unbiased if its expected value equals the true population parameter (θ).
Example: $\bar{Y}$ is an estimator for μ_Y , and it is unbiased because $\mathbb{E}[\bar{Y}] = \mu_Y$.

3. Law of Large Numbers (LLN): If 1) $Y_1, \dots, Y_n$ are iid, and 2) $\mathbb{E}[Y_i] < \infty$, then $\bar{Y} \xrightarrow{P} \mu_Y$.

4. Consistency: An estimator ($\hat{\theta}$) is consistent if the probability that it falls within any small interval of the true population parameter (θ) tends to one as the sample size increases. The following two statements are equivalent:

- a) $\hat{\theta}$ is a consistent estimator of θ .
- b) $\hat{\theta} \xrightarrow{P} \theta$.

The Central Limit Theorem says that

- a) $\bar{Y}$ is an unbiased estimator of μ_Y .
- b) $\bar{Y}$ has a standard normal distribution in large samples.
- c) $\bar{Y}$ is a consistent estimator of μ_Y .
- d) the distribution of $\bar{Y}$ is approximately normally distributed in large samples.

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Answer: d

$\bar{Y}$ is an *unbiased* estimator of the true population mean μ_Y means that

- a) the probability that $\bar{Y}$ is close to μ_Y increases as the sample size increases.
- b) the mean of the sampling distribution of $\bar{Y}$ is zero.
- c) the expected value of $\bar{Y}$ is equal to the true population mean μ_Y .
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Answer: c

1. Definitions: D_i is a treatment indicator variable.

Y_i^1 and Y_i^0 is the outcome in counterfactual worlds. Y_i^1 is the outcome if you take the treatment. Y_i^0 is the outcome if you do not take treatment.

In practice, we do not observe both counterfactual worlds at the same time – we cannot observe both Y_i^1 and Y_i^0 . We make a decision (to take treatment or not) and observe Y_i , an outcome, without a superscript.

$Y_i = Y_i^1$ if $D_i = 1$, $Y_i = Y_i^0$ if $D_i = 0$.

Hint: $\mathbb{E}[Y_i^1|D_i = 1]$ and $\mathbb{E}[Y_i^0|D_i = 0]$ is observed.

$\mathbb{E}[Y_i^1|D_i = 0]$ and $\mathbb{E}[Y_i^0|D_i = 1]$ is NOT observed.

2. Average treatment effect:

$$ATE = \mathbb{E}[Y_i^1 - Y_i^0].$$

3. Average treatment effect on the treated:

$$ATT = \mathbb{E}[Y_i^1 - Y_i^0|D_i = 1].$$

4. Observed difference is ATT + selection bias:

Observed difference in the population: $\mathbb{E}[Y_i^1|D_i = 1] - \mathbb{E}[Y_i^0|D_i = 0]$

$$\begin{aligned}\mathbb{E}[Y_i^1|D_i = 1] - \mathbb{E}[Y_i^0|D_i = 0] \\&= \mathbb{E}[Y_i^1|D_i = 1] - \mathbb{E}[Y_i^0|D_i = 1] + \mathbb{E}[Y_i^0|D_i = 1] - \mathbb{E}[Y_i^0|D_i = 0] \\&= \underbrace{\mathbb{E}[Y_i^1 - Y_i^0|D_i = 1]}_{ATT} + \underbrace{\mathbb{E}[Y_i^0|D_i = 1] - \mathbb{E}[Y_i^0|D_i = 0]}_{\text{selection bias}}\end{aligned}$$

Example

Consider two surgery patients, A and B. They can either receive an excellent surgeon or an average surgeon. Let $D_i = 1$ if the patient gets an excellent surgeon, $D_i = 0$ if they get an average surgeon, and let Y_i = amount of time in minutes the operation takes. Suppose:

	Patient A	Patient B
Y_i^0	60	80
Y_i^1	30	100
Y_i	??	??
D_i	0	1

For this population of patients, compute the ATE and ATT. Why might the sign be different?

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For this population of patients, compute the ATE and ATT. Why might the sign be different?

We have

$$ATE = \mathbb{E}[Y_i^1 - Y_i^0] = \frac{(Y_A^1 - Y_A^0) + (Y_B^1 - Y_B^0)}{2} = \frac{100 - 80}{2} + \frac{30 - 60}{2} = -5.$$

Also, only patient B is treated and only patient A is untreated, so

$$ATT = \mathbb{E}[Y_i^1 - Y_i^0 | D_i = 1] = \frac{Y_B^1 - Y_B^0}{1} = \frac{100 - 80}{1} = 20$$

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The sign might be different because the hospital might make the excellent surgeon take more complicated operations that take longer.

1. Write down the null and the alternative hypothesis:

$H_0 : \theta_1 = a$ and $H_1 : \theta_1 \neq a$, where a is some number (i.e. $a = 0$).

Note: The hypothesis is always about the population parameter, not the estimators!

2. Compute t-statistic:

$$t = \frac{\hat{\theta}_1 - a}{SE(\hat{\theta}_1)} = \frac{\hat{\theta}_1 - a}{\sqrt{\hat{\sigma}_{\hat{\theta}_1}^2}}$$

3. Compare t-statistic to critical values / calculate the p-values.
4. Use result from step 3 to either reject or fail to reject the null hypothesis.

Table 1: Trends in hourly earnings in the US of Working College Graduates, Age 25-34, 1996 to 2015, in 2015 dollars

Year	Male			Female		
	$\bar{Y}_m$	sY_m	n_m	$\bar{Y}_f$	sY_f	n_f
1996	24.87	11.44	1387	20.99	8.93	1232
2008	27.94	13.88	1855	23.16	11.11	1877
2015	28.06	14.37	1917	23.04	11.22	1816

Example (SW page 80)

1) Set up the null hypothesis:

$$H_0 : \mu_m - \mu_w = 0; \quad H_1 : \mu_m - \mu_w \neq 0 \quad (\text{two-sided})$$

2) Calculate the difference of averages and the standard error:

$$\bar{Y}_m - \bar{Y}_w = 28.06 - 23.04 = 5.02$$

and

$$SE(\bar{Y}_m - \bar{Y}_w) = \sqrt{\frac{s_m^2}{n_m} + \frac{s_w^2}{n_w}} = \sqrt{\frac{14.37^2}{1917} + \frac{11.22^2}{1816}} = 0.42$$

Note: Be very attentive of what the tables show you: s_{Y_m} or $s_{\bar{Y}_m}$?

3) Compute t-statistic: $\frac{\bar{Y}_m - \bar{Y}_w - 0}{SE(\bar{Y}_m - \bar{Y}_w)} = \frac{5.02}{0.42} = 11.95$.

4) We can reject the null hypothesis that male and female wage in 2015 was equal at the 1% significance level.

Linear Regression: $Y_i = \beta_0 + \beta_1 X_i + u_i, (i = 1, \dots, N)$

1. Y_i = dependent variable; X_i = independent variable; u_i = error term.
2. β_0 and β_1 are population parameters (intercept and slope, respectively).
Goal: identify relationship between X and Y .
3. We estimate β_0, β_1 with OLS (ordinary least squares):

$$\begin{aligned} \min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2 \\ \Rightarrow b_1 = \hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\widehat{\text{cov}}(X, Y)}{\widehat{\text{var}}(X)} \\ b_0 = \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \end{aligned}$$

4. Key OLS assumptions:

- (i) $\mathbb{E}[u_i | X_i = x] = 0$.
- (ii) X_i and Y_i are iid (identically and independently distributed) $\forall i$.
- (iii) Large outliers are rare ($\mathbb{E}[X^4] < \infty$, $\mathbb{E}[Y^4] < \infty$).

5. The standard error of $\hat{\beta}_1$ gets smaller as $n \uparrow$ and as $\sigma_X^2 \uparrow$

Notes:

$(\hat{\beta}_0, \hat{\beta}_1)$ are consistent. In large samples, CLT applies to $\hat{\beta}$.

$\hat{u}_i = Y_i - \hat{Y}_i$ is the residual; u_i is the error term.

$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ is the predicted value.

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$$

R^2 = fraction of variation in Y that is explained by X ; unitless.

$$R^2 = \frac{ESS}{TSS} = \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

SER = standard error of the regression: measure of the magnitude of the typical regression residual in the same units as Y .

Be careful about units of $\hat{\beta}_0$ and $\hat{\beta}_1$ (and units of X and Y).